

MAK HIVOLTOL R

High performance uninhibited electrical insulating oil

MAK Hivoltol R is a high quality mineral uninhibited electrical insulating oil. It is manufactured from severely hydrotreated low aromatic oil and does not have polar compounds. It has very good dielectric property and high resistance to oxidation coupled with low temperature property and efficient heat transfer. MAK Hivoltol R has a high resistance to thermal and chemical degradation and offers very high breakdown voltage. It is non-corrosive towards copper and does not require passivation.

Applications:

MAK Hivoltol R is recommended to be used as an electrical insulation medium in oil filled grid and industrial transformers.

MAK Hivoltol R is not suitable for use in oil filled cables or for special impregnation processes.

Performance/ Benefits:

Excellent Thermal and Chemical Stability – high resistance to thermal and chemical degradation in presence of iron and copper. Reduces tendency to produce sludge and oil soluble oxidation products. Enhances operating life.

Very High Dielectric Strength – allows the oil to withstand electrical stress. Indicates the absence of oil contaminants like moisture, fibres and polar chemicals.

Extended Oil Life – offers inherent natural resistance to oxidation of oil and allows longer working life. Enhances equipment reliability.

Extremely Low Water Content – specially treated to achieve an extremely low water content and retain very high break down voltage. Offers high resistance to oxidation of oil.

High System Efficiency – circulates freely at low temperatures and ensures rapid heat transfer in transformers. It is able to quench arcs quickly in switchgear units. Maintains reliability of tap changers at low temperatures.

Specification:

- IS 335:2018 (Type II) – Uninhibited

Typical Physico-Chemical Data: MAK Hivoltol R

Characteristics	Method	Value
Appearance	Visual	Clear
Density, g/cc @20°C	IS:1448 (P-16)	0.895, max
Kinematic Viscosity @0°C, cSt	IS:1448 (P-25)	1800, max
Kinematic Viscosity @40°C, cSt	IS:1448 (P-25)	15.0, max
Flash Point, PMCC, °C	IS:1448 (P-21)	140, min
Pour Point, °C	IS:1448 (P-10)	-10
Acidity, mg KOH/ g	IEC 62021-1	0.01, max
Corrosive Sulphur	DIN 51353	NC
Potential Corrosive Sulphur	IS: 16310	NC
Electric Strength (Break Down Voltage)	IS: 6792	
New Unfiltered Oil, kV		30, min
After filtration		70, min
Dielectric Dissipation Factor (tan δ) @90°C	IS: 16086	0.005, max
Interfacial Tension, mN/ m	ASTM D971	40, min
DBDS	IS: 16497 (Part 1)	ND
Metal Passivator Additives	IS: 13631	ND
Oxidation Stability	IS: 12422, Method C, 164 hrs.	
a) Total Acidity, mg KOH/g		1.2, max
b) Total Sludge after oxidation, % by wt.		0.8, max
c) DDF at 90°C		0.5, max
Water Content, ppm	IEC 60814	30, max
PCA, %	IP: 346	3, max
PCB, ppm	IS: 16082	ND
Presence of Oxidation Inhibitor	IS: 13631	ND

ND – Not Detectable, NC – Non-Corrosive

Storage & Handling:

Trace amount of contamination with foreign materials can severely endanger the critical properties of the oil. Commonly encountered contaminants are moisture, fibres, particles and surfactants. This product should be kept clean and dry and be stored indoor. Keep it properly sealed to avoid contamination. Shelf life is 3 yrs. under protected storage conditions.

Health & Safety:

It is unlikely to be hazardous when properly used in recommended applications. Contamination of the oil from moisture, fibres, particles and surfactants can occur during the use. It should be avoided. It is free from polychlorinated biphenyls (PCB) and polycyclic aromatics (PCA). Regular monitoring of the in-use product is recommended.